Singapore School Kinderland

Secondary 1 (Sec 1) Computer Science — Term 2 Comprehensive Study Material

Prescribed Textbook: Cambridge IGCSE & O Level Computer Science (Second Edition) | Chapter 2: Data Transmission (Pages 54–74)

PART 1: CHAPTER 2.2 — METHODS OF ERROR DETECTION (TB Pages 54–62)

During data transmission across communication networks, electrical interference, power surges, signal attenuation, or packet collisions can cause binary bits to flip ($0 \rightarrow 1$ or $1 \rightarrow 0$). Computer scientists use mathematical error detection methods to identify corrupted packets.

1. Parity Checks (Odd & Even Parity) — TB Pages 54–55

A byte is made of 8 bits: 7 bits of actual data and 1 **Parity Bit** allocated to ensure the total number of 1s in the byte conforms to an established parity rule.

- **Even Parity:** The parity bit is set to 0 or 1 so that the total count of 1s in the 8-bit byte is an **EVEN** number (0, 2, 4, 6, 8).
- **Odd Parity:** The parity bit is set so that the total count of 1s in the byte is an **ODD** number (1, 3, 5, 7).

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Example (Even Parity): Data = 1 0 1 1 0 0 1 (Count of 1s = 4, already even) →
Parity Bit = 0 → Byte: 0 1 0 1 1 0 0 1
Example (Odd Parity): Data = 1 0 1 1 0 0 1 (Count of 1s = 4, currently even) →
Parity Bit = 1 → Byte: 1 1 0 1 1 0 0 1
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> **Cambridge Exam Limitation:** A simple parity check cannot detect errors if an **even number of bits** flip in the same byte (e.g. two 0s flip to 1s). The total count of 1s remains even/odd, causing the corruption to go completely undetected!

2. Parity Blocks / Parity Byte Matrix — TB Pages 56–57

To overcome single-byte limitations, data bytes are assembled into a 2D block (matrix). Horizontal parity bits are appended to every row, and an extra vertical **Parity Byte** is added at the bottom.

When transmitted, if a bit flips, both its horizontal row parity and its vertical column parity will show an error. The intersection of the corrupted row and corrupted column pinpoints the exact bit that flipped, enabling **automatic error correction** without retransmission!